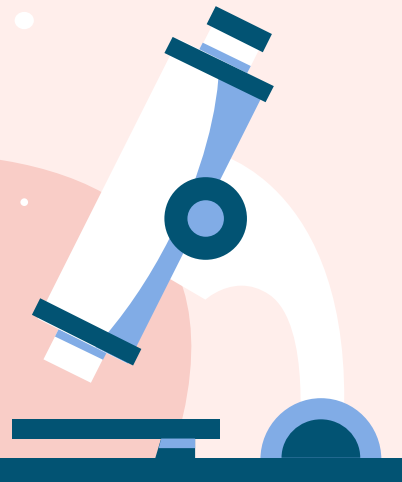
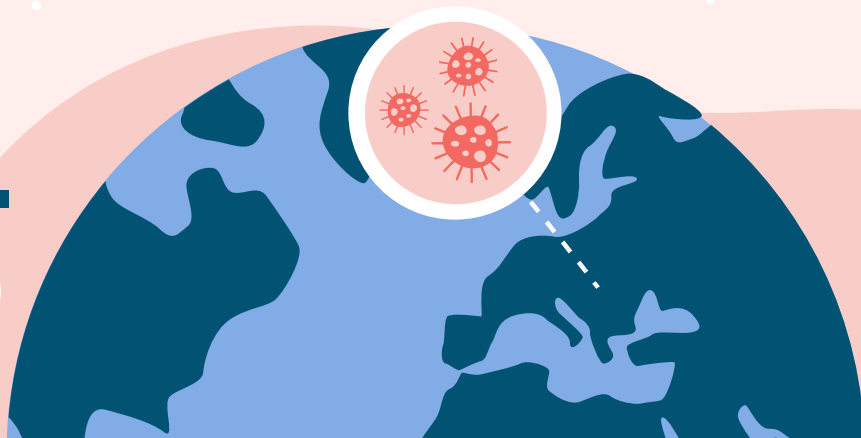
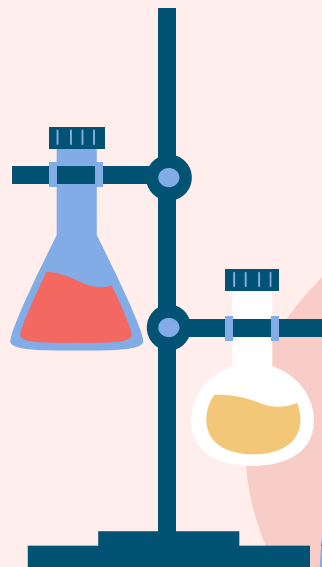


Delineo

Use Cases

Xuanyi Wang, Jayden Zhang





Delineo Use case ideas

POI hotspots

Which schools, shops and venues drive spread (our example)

Targeted closures

Close specific venues and compare with a full lockdown

Intervention timing

How much the start hour changes outbreak size

Who gets sick

Break outcomes down by age and sex

Hospital planning

Forecast hospitalized cases over time

Town comparison

Run the same scenario in different communities

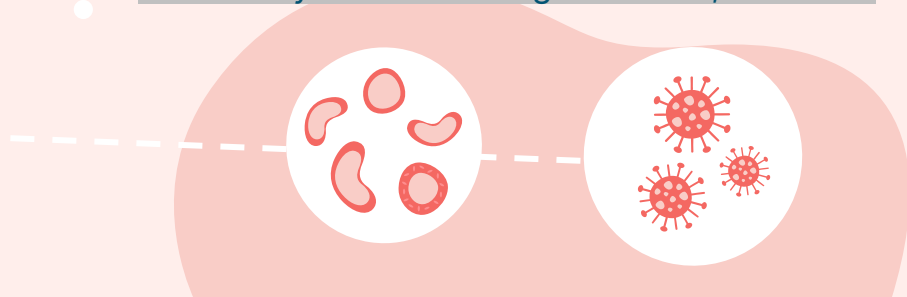
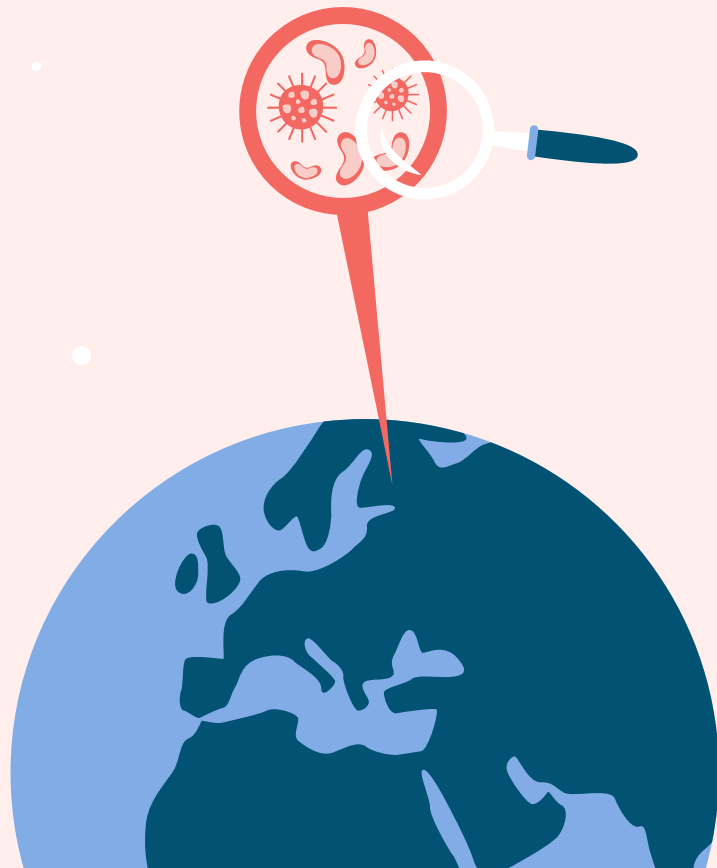


01

POI hotspots

Barnsdall, OK

Note: analyze data with a grain of salt; first draft.





Ten scenarios



Baseline

B0: no intervention, one initial infected person

Single measures

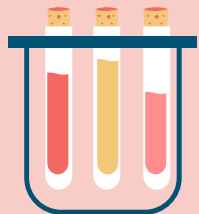
50% masking, vaccination, lockdown or self-isolation from hour 12

Combinations

Masking + vaccination, lockdown + isolation, or all four at hour 12

Vaccine timing

50% vaccination starting at hour 12, 72 or 168





Analysis pipeline

Load

Chart and map data,
cached offline via API

Compare

Infections across 3,180
POIs

01

02

03

04

05

Register

Run IDs and scenario
labels; list them

Audit

Double checking
algorithm; Compares
server side with inputs

Visualize

Six figures and CSV
tables (jupyter nb)





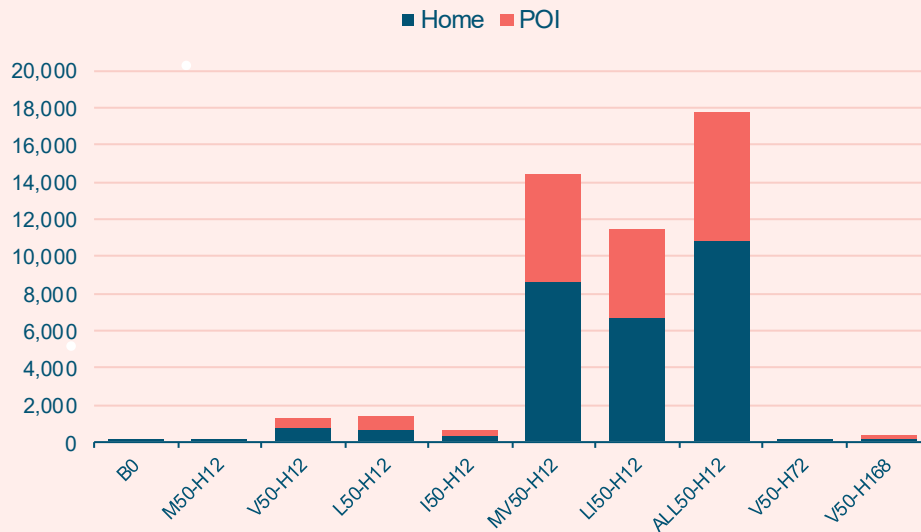
Where infections happened

Combined scenarios (MV, LI, ALL) show the largest outbreaks.

When outbreaks grow, 39–51% of infections happen at POIs.

B0, M50-H12 and V50-H72 did not manage to spread.

Attributed infections by scenario



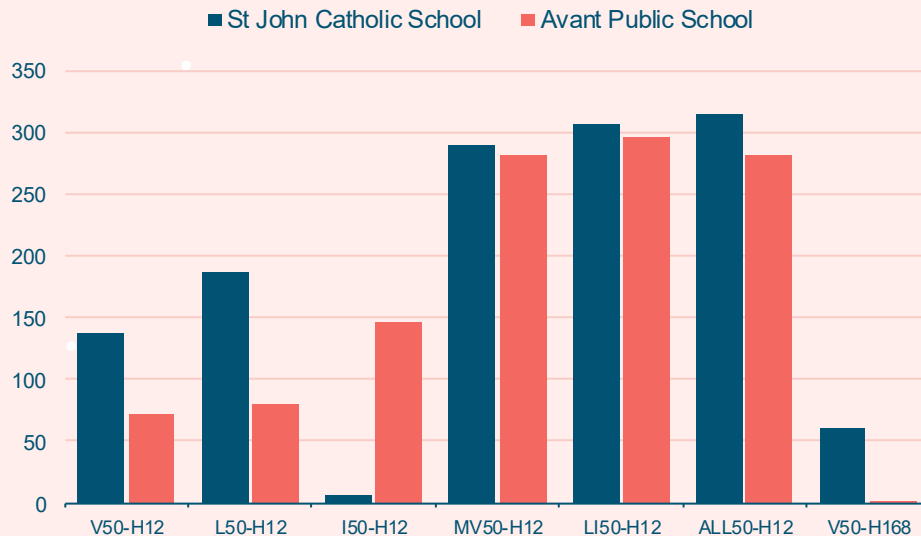
POI spread and, ironically, schools

St John Catholic School is the #1 hotspot in 6 of the 7 scenarios that spread.

Avant Public School ranks #1 or #2 in 6 of 7.

As such, **schools** are the top POI category regarding spread in every scenario.

Infections at the two focal schools



Future steps/implications



Target schools

Test masking or closures at St John and Avant alone or other interventions



Does time matter?

Compare different start hours to see if timing matters and how much it does



Repeat runs

We used random seeds for the runs, so repeating will give us better results

Thanks!

Do you have any questions?



CREDITS: This presentation template was created by **Slidesgo**, and includes icons by **Flaticon** and infographics & images by **Freepik**

